in the same grid are mapped to the same output token at each step. Then for any input $\mathbf{x} \in [0,1]^{d\times m_1}$ and any f^j, we can find a prompt $\mathbf{P}_j$ such that

$$
\begin{aligned}
\|\tau([\mathbf{P}_j, \mathbf{x}])_{:,-1} - f_0^j(\mathbf{x})\|_p &\leq m_2^{-1/p}\epsilon \\
\|\tau([\mathbf{P}_j, \mathbf{x}, \tau([\mathbf{P}, \mathbf{x}])_{:,-1}]_{:,-1}, f_1^j([\mathbf{x}, \tau([\mathbf{P}, \mathbf{x}])_{:,-1}])\|_p &\leq m_2^{-1/p}\epsilon \\
&\dots
\end{aligned}
$$

Then we have $d_p(h(\mathbf{P}_j), f^j) \leq \epsilon$. □

C.5 Proof of Lemma 4

If $\|\mathbf{W}_1\|_2 \times \|\mathbf{W}_2\|_2 < 1$, where $\|\cdot\|_2$ is the matrix spectral norm, then the MLP block in Definition 2 is invertible, ie, $\texttt{MLP}^{-1}$ is a singleton set.

Proof. Based on the sufficient conditions for invertibility of a residual block Behrmann et al. [2019], we have that if the feedforward part of a residual block $\mathbf{W}_2\texttt{ReLU}(\mathbf{W}_1\mathbf{X}_{:,1} + \mathbf{b}_1) + \mathbf{b}_2$ is a contraction with respect to some metric, i.e. its Lipschitz constant < 1, and the metric space on which is defined is complete, then MLP in eq 2 is invertible. Since we are dealing with the euclidean space, any metric induced by the $\|\cdot\|_p$ norm for $p \in [1, \infty]$ ensures the space is complete.
The Lipschitz constant of $\mathbf{W}_2\texttt{ReLU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$ is simply $\|\mathbf{W}_1\|_2 \times \|\mathbf{W}_2\|_2$. Thus the statement of the lemma follows. □

C.6 Proof of Theorem 2

For a single layer transformer τ defined above with Assumptions 1 and 2, we can build a seq-to-seq dataset $\{(\mathbf{X}_1 = [\mathbf{x}_1, \mathbf{x}_0], \mathbf{Y}_1 = [\mathbf{y}_{11}, \mathbf{y}_{10}]), (\mathbf{X}_2 = [\mathbf{x}_2, \mathbf{x}_0], \mathbf{Y}_2 = [\mathbf{y}_{21}, \mathbf{y}_{20}]))\}$, and we cannot find a prompt $\mathbf{P} \in \mathbb{R}^{d\times m_p}$ with any $m_p > 0$ such that $\tau([\mathbf{P}, \mathbf{X}_i]) = \mathbf{Y}_i$ holds for any $i = 1, 2$. The vectors $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2$ are denoted post positional encodings.

Proof. Before proving Theorem 2, we first provide a lemma that will be used in proof and also Theorem 3.

Lemma 7. *Given any $\mathbf{c} \in \mathbb{R}^{d\times m}$, there are $\mathbf{x}_0$ almost anywhere for which we can find another vector $\mathbf{x}_1 \in \mathbb{R}^{d\times m}$ such that $\texttt{Att}(\mathbf{x}_0, [\mathbf{x}_0, \mathbf{x}_1]) \parallel \mathbf{c}$ with full rank attention weights $\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v$.*

Proof. If $\mathbf{W}_v\mathbf{x}_0 \parallel \mathbf{c}$, we can just set $\mathbf{x}_1 = \mathbf{x}_0$, which makes $\texttt{Att}(\mathbf{x}_0, [\mathbf{x}_0, \mathbf{x}_1]) \parallel \mathbf{c}$ hold.

If $\mathbf{W}_v\mathbf{x}_0 \nparallel \mathbf{c}$, let $\mathbf{v} = \alpha\mathbf{c} - \mathbf{W}_v\mathbf{x}_0$ where $\alpha \in \mathbb{R}$. As $\mathbf{W}_v$ is full-rank, we can find $\mathbf{x}$ such that $\mathbf{x} = \mathbf{W}_v^{-1}\mathbf{v} = \alpha\mathbf{W}_v^{-1}\mathbf{c} - \mathbf{x}_0$. Then we will have

$$
\begin{aligned}
&\texttt{Att}(\mathbf{x}_0, [\mathbf{x}_0, \mathbf{x}_1]) \\
=&\frac{\exp\left((\mathbf{W}_q\mathbf{x}_0)^\top(\mathbf{W}_k\mathbf{x}_0)\right)\mathbf{W}_v\mathbf{x}_0 + \exp\left((\mathbf{W}_q\mathbf{x}_0)^\top(\mathbf{W}_k(\alpha\mathbf{W}_v^{-1}\mathbf{c} - \mathbf{x}_0))\right)(\alpha\mathbf{c} - \mathbf{W}_v\mathbf{x}_0)}{\exp\left((\mathbf{W}_q\mathbf{x}_0)^\top(\mathbf{W}_k\mathbf{x}_0)\right) + \exp\left((\mathbf{W}_q\mathbf{x}_0)^\top(\mathbf{W}_k(\alpha\mathbf{W}_v^{-1}\mathbf{c} - \mathbf{x}_0))\right)}
\end{aligned}
$$

Therefore, as long as $\mathbf{W}_q\mathbf{x}_0 \not\perp \mathbf{W}_k(\mathbf{W}_v^{-1}\mathbf{c})$, we can change α such that $\texttt{Att}(\mathbf{x}_0, [\mathbf{x}_0, \mathbf{x}_1]) = \beta\mathbf{W}_v\mathbf{x}_0 + (1-\beta)(\alpha\mathbf{c} - \mathbf{W}_v\mathbf{x}_0)$ where $\beta = \frac{\exp\left((\mathbf{W}_q\mathbf{x}_0)^\top(\mathbf{W}_k\mathbf{x}_0)\right)}{\exp\left((\mathbf{W}_q\mathbf{x}_0)^\top(\mathbf{W}_k\mathbf{x}_0)\right) + \exp\left((\mathbf{W}_q\mathbf{x}_0)^\top(\mathbf{W}_k(\alpha\mathbf{W}_v^{-1}\mathbf{c} - \mathbf{x}_0))\right)}$. When $\alpha = 0$, $\texttt{Att}(\mathbf{x}_0, [\mathbf{x}_0, \mathbf{x}_1]) = \mathbf{W}_v\mathbf{x}_0$, when $\alpha \to -\infty$ or $\alpha \to \infty$, $\texttt{Att}(\mathbf{x}_0, [\mathbf{x}_0, \mathbf{x}_1]) \to \alpha\mathbf{c} - \mathbf{W}_v\mathbf{x}_0$. As $\texttt{Att}(\mathbf{x}_0, [\mathbf{x}_0, \mathbf{x}_1])$ is continuous w.r.t changing α, there must exist an α such that $\texttt{Att}(\mathbf{x}_0, [\mathbf{x}_0, \mathbf{x}_1]) \parallel \mathbf{c}$. □

Pass the two input sequences $\mathbf{X}_1, \mathbf{X}_2$ through the attention layer Att with any prompt $\mathbf{P}$, we can get the last output token as:

$$
\texttt{Att}(\mathbf{x}_0, [\mathbf{P}, \mathbf{X}_1]) = \lambda(\mathbf{X}_1, \mathbf{x}_0, [\mathbf{P}, \mathbf{X}_1])\texttt{Att}(\mathbf{x}_0, \mathbf{X}_1]) + \lambda(\mathbf{P}, \mathbf{x}_0, [\mathbf{P}, \mathbf{X}_1])\texttt{Att}(\mathbf{x}_0, \mathbf{P}) \tag{12}
$$

$$
\texttt{Att}(\mathbf{x}_0, [\mathbf{P}, \mathbf{X}_2]) = \lambda(\mathbf{X}_2, \mathbf{x}_0, [\mathbf{P}, \mathbf{X}_2])\texttt{Att}(\mathbf{x}_0, \mathbf{X}_2) + \lambda(\mathbf{P}, \mathbf{x}_0, [\mathbf{P}, \mathbf{X}_2])\texttt{Att}(\mathbf{x}_0, \mathbf{P}) \tag{13}
$$

Here $\lambda(\mathbf{X}_1, \mathbf{x}_2, \mathbf{X}_3 = [\mathbf{X}_1, \mathbf{X}_2]) \in (0, 1)$ is a positive scalar, defined as

$$\lambda(\mathbf{X}_1, \mathbf{x}_2, \mathbf{X}_3) = \frac{\sum_j \exp((\mathbf{W}_k \mathbf{x}_{1j})^\top (\mathbf{W}_q \mathbf{x}_2))}{\sum_j \exp((\mathbf{W}_k \mathbf{x}_{3j})^\top (\mathbf{W}_q \mathbf{x}_2))}.$$

$\mathbf{x}_{ij}$ is the jth token in $\mathbf{X}_i$ for notation simplicity.

1. Then from equation 12, $\mathtt{Att}(\mathbf{x}_0, \mathbf{P})$ must be on $\mathtt{Cone}(-\mathtt{Att}(\mathbf{x}_0, \mathbf{X}_1), \mathtt{Att}(\mathbf{x}_0, [\mathbf{P}, \mathbf{X}_1]))$ and $\mathtt{Cone}(-\mathtt{Att}(\mathbf{x}_0, \mathbf{X}_2), \mathtt{Att}(\mathbf{x}_0, [\mathbf{P}, \mathbf{X}_2]))$.
2. On the otherhand, as we want to memorize the two examples, we must have $\mathtt{Att}(\mathbf{x}_0, [\mathbf{P}, \mathbf{X}_1]) + \mathbf{x}_0 \in \mathtt{MLP}^{-1}(\mathbf{y}_{10})$ and $\mathtt{Att}(\mathbf{x}_0, [\mathbf{P}, \mathbf{X}_2]) + \mathbf{x}_0 \in \mathtt{MLP}^{-1}(\mathbf{y}_{20})$.

We construct the dataset S with arbitrary $\mathbf{x}_0, \mathbf{y}_{10}$ and $\mathbf{y}_{20}$. Then if $\dim((\mathtt{MLP}^{-1}(\mathbf{y}_{10}) - \mathbf{x}_0) \cup (\mathtt{MLP}^{-1}(\mathbf{y}_{20}) - \mathbf{x}_0)) + 2 \leq d$ (Assumption 2), we can find two vectors $\mathbf{c}_1, \mathbf{c}_2$ such that $\mathbf{c}_1, \mathbf{c}_2 \perp \mathbf{v} : \mathbf{v} + \mathbf{x}_0 \in \mathtt{MLP}^{-1}(\mathbf{y}_{10})$ or $\mathbf{v} + \mathbf{x}_0 \in \mathtt{MLP}^{-1}(\mathbf{y}_{20})$ and $\mathbf{c}_1 \perp \mathbf{c}_2$. Then we can choose $\mathbf{x}_1, \mathbf{x}_2$ such that $\mathtt{Att}(\mathbf{x}_0, \mathbf{X}_1) \parallel \mathbf{c}_1$ and $\mathtt{Att}(\mathbf{x}_0, \mathbf{X}_2) \parallel \mathbf{c}_2$ (Lemma 7). Combine this construction with assumption 1, we have that $\mathtt{Cone}(-\mathtt{Att}(\mathbf{x}_0, \mathbf{X}_1), \mathtt{Att}(\mathbf{x}_0, [\mathbf{P}, \mathbf{X}_1]))$ and $\mathtt{Cone}(-\mathtt{Att}(\mathbf{x}_0, \mathbf{X}_2), \mathtt{Att}(\mathbf{x}_0, [\mathbf{P}, \mathbf{X}_2]))$ has no intersection, which means that we cannot find a $\mathbf{P}$ to memorize this constructed dataset. □

C.7 Proof of Lemma 5

For a standard single-layer transformer τ defined in Definition 2 with $r \geq n$ MLP hidden neurons, for any sequence-to-sequence dataset S satisfying Assumptions 1, we can apply a low-rank update to MLP weights with $O(nd)$ parameters to memorize $\tau(\mathbf{X}_i)_{:,m} = \mathbf{y}_{im}$.

Proof. We use $\mathtt{MLP}_j(\mathbf{x})$ to denote the jth output of the MLP layer for an input token $\mathbf{x}$, which is

$$\mathtt{MLP}_j(\mathbf{x}) = x_j + b_{2,j} + \sum_{k=1}^{m} w_{k,j} \max(\langle \mathbf{a}_k, \mathbf{x} \rangle + b_{1,k}, 0)$$

According to our assumption, $\mathtt{Att}(\mathbf{x}_{im}, \mathbf{X}_i)$ are unique vectors for $i = 1, 2, ..., n$. Then we only need to use the MLP layer to map each $\mathbf{x}_i = \mathtt{Att}(\mathbf{x}_{im}, \mathbf{X}_i) + \mathbf{x}_{im}$ to $\mathbf{y}_{im}$, where we get a new token-wise dataset $\{(\mathbf{x}_1, \mathbf{y}_1), (\mathbf{x}_2, \mathbf{y}_2), ..., (\mathbf{x}_n, \mathbf{y}_n)\}$

Then we need to find w_k, $\mathbf{a}_k$ and b_k such that

$$\mathtt{MLP}_j(\mathbf{x}_i) = x_{i,j} + b_{2,j} + \sum_{k=1}^{m} w_{k,j} \max(\langle \mathbf{a}_k, \mathbf{x}_i \rangle + b_{1,k}, 0) = y_{i,j}, i = 1, 2, ..., n, j = 1, 2, ..., d \tag{14}$$

, which is equivalent to constructing a standard MLP to memorize a dataset:

$$\sum_{k=1}^{n} w_{k,j} \max(\langle \mathbf{a}_k, \mathbf{x}_i \rangle + b_{1,k}, 0) = y_{i,j} - x_{i,j} - \sum_{k=n+1}^{m} w_{k,j} \max(\langle \mathbf{a}_k, \mathbf{x}_i \rangle + b_{1,k}, 0) - b_{2,j} \tag{15}$$

Follow Thoerem 1 in Yun et al. [2019b], we can construct $\mathbf{a}, b_1, ..., b_n$ such that for $\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_n$, we have $z_i = \langle \mathbf{a}, \mathbf{x}_i \rangle$, $b_1 < z_1 < b_2 < ... < b_n < z_n$. Then we can find $w_1, ..., w_n$ which solves equation 15. For d-dimension output, we need to find $\mathbf{W} \in \mathbb{R}^{n \times d}$ and $\mathbf{a} \in \mathbb{R}^d$ and $\mathbf{b} \in \mathbb{R}^n$. With LoRA, we need a low-rank update of size $m \times n + n \times d$ for $\mathbf{W}_2$, a low-rank update of size $d \times n + n \times m$ for $\mathbf{W}_1$ and an update of size n for $\mathbf{b}_1$, which is $O(n \times d)$ in total. Normally we have $m \simeq d$, then we need an update with parameter size around $(4n + 1)d$ to memorize the last token of n training examples. □

C.8 Proof of Theorem 3

For any single layer transformer τ defined in Definition 2, there exists a seq-to-seq dataset $\{(\mathbf{X}_1 = [\mathbf{x}_{10}, \mathbf{x}_1], [\mathbf{y}_{10}, \mathbf{y}_{11}]), (\mathbf{X}_2 = [\mathbf{x}_{20}, \mathbf{x}_2], [\mathbf{y}_{20}, \mathbf{y}_{21}]), ..., (\mathbf{X}_n = [\mathbf{x}_{n0}, \mathbf{x}_n], [\mathbf{y}_{n0}, \mathbf{y}_{n1}])\}$ that satisfies Assumption 1 with $n < d$ training examples such that we need at least n prompt tokens in $\mathbf{P}$ to memorize the training set, ie, for $\tau([\mathbf{P}, \mathbf{X}_i])_{:,-1} = \mathbf{y}_{i1}$ to hold for all $i = 1, 2, ..., n$.

Proof. Without loss of generality, we assume $\mathbf{W}_2$ has no zero elements, otherwise we can just ignore this hidden neuron in MLP layer.

$\mathbb{R}^d$ has d bases $\{\mathbf{t}_j : j = 1, 2, ..., d\}$, then $\texttt{MLP}^{-1}(\mathbf{y}_{i1})$ must be bounded on either positive or negative part of these d directions, which means there exists $B \geq 0$ such that

$$\frac{\mathbf{v}^\top \mathbf{t}_j}{\|\mathbf{t}_j\|} \leq B, \forall \mathbf{v} \in \texttt{MLP}^{-1}(\mathbf{y}_{i1}), j = 1, 2, ..., d$$

Otherwise $\forall B > 0$, $\exists \mathbf{t}_j$, we can find a $\mathbf{v} \in \texttt{MLP}^{-1}(\mathbf{y}_i)$ that $\frac{\mathbf{v}^\top \mathbf{t}_j}{\|\mathbf{t}_j\|} \geq B$. Meanwhile we have $\texttt{MLP}(\mathbf{v}) = \mathbf{v} + \mathbf{b}_2 + \mathbf{W}_2 \texttt{ReLU}(\mathbf{W}_1 \mathbf{v} + \mathbf{b}_1)$. As $\|\mathbf{v}\|$ can be arbitrarily large, if $\mathbf{W}_1 \mathbf{v} = \mathbf{0}$, $\|\texttt{MLP}(\mathbf{v})\| \to \infty$ if $\|\mathbf{v}\| \to \infty$. if $\mathbf{W}_1 \mathbf{v} \neq \mathbf{0}$, $\|\texttt{MLP}(\mathbf{v})\|$ can also be arbitrarily large when increasing the norm of $\mathbf{v}$ due to the non-linearity of $\texttt{ReLU}(\mathbf{W}_1 \mathbf{v} + \mathbf{b}_1)$.

Then we can find a set of n linearly independent vectors $\{\mathbf{c}_1, \mathbf{c}_2, ..., \mathbf{c}_n\}$ such that $\{\mathbf{a}_i : \mathbf{a}_i - \mathbf{c}_i \perp \mathbf{c}_i, \mathbf{a}_i \in \texttt{MLP}^{-1}(\mathbf{y}_{i1})\} = \emptyset$ by enlarging the norm of $\mathbf{c}_i$. With the n $\mathbf{c}_i$ vectors, we can begin to construct our dataset:

We set $\mathbf{x}_i = \mathbf{c}_i, i = 1, 2, ..., n$ and find $\mathbf{x}_{i0}$ such that $\mathbf{c}_i \perp \texttt{Att}(\mathbf{x}_i, \mathbf{X}_i)$ (Lemma 7) and $\texttt{Att}(\mathbf{x}_i, \mathbf{X}_i)$ are distinct for $i = 1, 2, ..., n$ (Assumption 1), which makes $\{\mathbf{a}_1 - \mathbf{x}_1 - \lambda(\mathbf{X}_1, \mathbf{x}_1, [\mathbf{P}, \mathbf{X}_1])\texttt{Att}(\mathbf{x}_1, \mathbf{X}_1), ..., \mathbf{a}_n - \mathbf{x}_n - \lambda(\mathbf{X}_n, \mathbf{x}_n, [\mathbf{P}, \mathbf{X}_n])\texttt{Att}(\mathbf{x}_n, \mathbf{X}_n)\}$ linearly independent for any $\mathbf{a}_i \in \texttt{MLP}^{-1}(\mathbf{y}_{i1})$. Here $\lambda(\cdot, \cdot, \cdot)$ is the same as defined in Section C.6.

Moreover, we have

$$\begin{aligned} \texttt{Att}(\mathbf{x}_i, [\mathbf{P}, \mathbf{X}_i]) &= \lambda(\mathbf{X}_i, \mathbf{P}, [\mathbf{P}, \mathbf{X}_i])\texttt{Att}(\mathbf{x}_i, \mathbf{P}) + \lambda(\mathbf{X}_i, \mathbf{x}_i, [\mathbf{P}, \mathbf{X}_i])\texttt{Att}(\mathbf{x}_i, \mathbf{X}_i) \\ &\in \texttt{MLP}^{-1}(\mathbf{y}_{i1}) - \mathbf{x}_i \end{aligned} \tag{16}$$

Then $\texttt{Att}(\mathbf{x}_i, \mathbf{P}), i = 1, 2, ..., n$ must be n linearly independent vectors, which requires

$$\texttt{rank}(\mathbf{W}_v \mathbf{P} \mathbf{A}) = n, \tag{17}$$

where $\mathbf{A} \in \mathbb{R}^{m_p \times n}$ is the attention score matrix between $\mathbf{x}_i$ and $\mathbf{P}$. $\mathbf{P} \in \mathbb{R}^{d \times m_p}$ is the prompt token sequence and $\mathbf{W}_v$ is the attention value weight. Therefore, we must have $m_p \geq n$. □

C.9 Proof of Theorem 4

A transformer $\mathcal{T}$ is invertible if:

1. The Lipschitz constant of the attention block in each layer τ^l is *strictly* less than 1
2. The Lipschitz constant of the 2-layer ReLU block in each layer τ^l, which is bounded by $\|\mathbf{W}_2^l\|_2 \times \|\mathbf{W}_1^l\|_2$, is *strictly* less than 1

Proof. This proof is based on the proof provided for lemma 4, thus we restrict ourselves to the sketch: Based on the sufficient condition for invertibility in Behrmann et al. [2019], condition (1) implies that the attention block (eq 1) with the residual connection, ie $\mathbf{X} + \texttt{Att}(\mathbf{X}, \mathbf{X})$, is an invertible function. Similarly, condition (2) implies that the MLP block which constitutes of the 2-layer ReLU block with the residual connection (eq 2) also exhibit invertibility.
Thus each transformer layer τ^l (eq 3) is invertible by noting that its a composition of two invertible functions. The same property ensures that the entire transformer architecture $\mathcal{T}$ is also invertible. □

C.10 Proof of Lemma 6

Under the compactness condition, the Lipschitz constant of the i-th attention head in the l-th transformer layer, denoted for simplicity as $\texttt{Att}^{i,l}$, admits the following bound w.r.t the entire input sequence of length m:

$$Lip(\texttt{Att}^{i,l}(\cdot, \cdot)) \leq (1 + 8\sqrt{m}(D^l)^2 \|(\mathbf{W}_k^{i,l})^T \mathbf{W}_q^{i,l}\|_2)\|\mathbf{W}_v^{i,l}\|_2, \tag{18}$$

and the Lipschitz constant of the entire attention block in layer l, denoted as $\texttt{Att}^l$, admits the bound:

$$Lip(\texttt{Att}^l(\cdot, \cdot)) \leq \sqrt{\sum_{i=1}^{h} (\|\mathbf{W}_o^{i,l}\|_2 \times Lip(\texttt{Att}^{i,l}))^2}. \tag{19}$$